

Question:
What is the de Broglie wavelength of an electron with a kinetic energy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

<think>

Use the de Broglie relation for an electron in free space : $\lambda = h/p$, where $p = \sqrt{2 m_e K}$ and K is the kinetic energy . For convenience , use $1 \text{ eV} = 1.602 \times 10^{-11} \text{ J}$ and electron charge e .

- Energy per mass : $1 \text{ eV} = E_e \approx 1.602 \times 10^{-11} \text{ J}$.
- Kinetic energy : $K_e = E_e = 1.602 \times 10^{-11} \text{ J}$.
- Momentum : $p = \sqrt{2 m_e K_e}$.
- Electron mass : $m_e \approx 9.109 \times 10^{-31} \text{ kg}$.
- Hence $p \approx \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-11}} \approx 7.32 \times 10^{-22} \text{ kg}\cdot\text{m/s}$.
- De Broglie wavelength : $\lambda = h/p \approx (6.626 \times 10^{-34} \text{ J}\cdot\text{s}) / (7.32 \times 10^{-22}) \approx 9.04 \times 10^{-13} \text{ m}$.

This corresponds to $\approx 9.81 \times 10^{-4} \text{ cm}$ (not $5.27 \times 10^{-9} \text{ m}$, which is a millimeter value) .

Correct answer : $9.81 \times 10^{-4} \text{ cm}$.

</think>

<answer>

$9.81 \times 10^{-4} \text{ cm}$

</answer>

